

Fixed clocks captured most of the measured savings

Adaptive clock control added little beyond a good fixed setting in these inference replays. Across eight paired groups, the selected demand policy used 0.77% less telemetry-derived GPU energy than a fixed 660 MHz request. The reduction was 1.17% on code and 0.29% on conversation. All 24 comparison trials completed, covering four unique Azure length-and-timing blocks, each repeated twice, on NVIDIA L4 allocations running Qwen2.5-7B-Instruct.

The baseline changes the apparent result. Adaptive control used 11.00% less energy than GPU-managed graphics clocks across the same groups. Fixed 660 MHz alone used 10.31% less than GPU-managed clocks. Most of that observed benefit was therefore available with a constant request. GPU-managed here retains the same configured 40 W limit and locked 6,251 MHz memory; it is not a fully stock device configuration.

The lowest clock was not the best fixed choice. On one 64-request code development block, raising the request from 210 to 660 MHz reduced energy by 26.37%. Average board power rose, but completion time fell from 238 to 144 seconds. Conversation made a different tradeoff: 660 MHz used 4.04% more energy than 210 MHz while substantially improving latency. Screening all five requested settings selected 660 MHz as the common lowest-energy setting meeting both sources' development thresholds. Those thresholds were benchmark-relative, set at 1.1 times the corresponding 660 MHz measurements.

Prefill and decode still help explain the workload, but requested frequency is only part of the story. Long-prompt, one-output-token probes used about 30-32% less whole-workload energy at 660 than at 210 MHz. Short-prompt, long-generation probes differed by less than 0.5%. Each concurrency had two paired repetitions. In singleton decode, the realized clock stayed near 210 MHz under either request. A higher requested clock therefore did not guarantee a faster physical operating point. These request-shape probes do not isolate exact GPU phase energy.

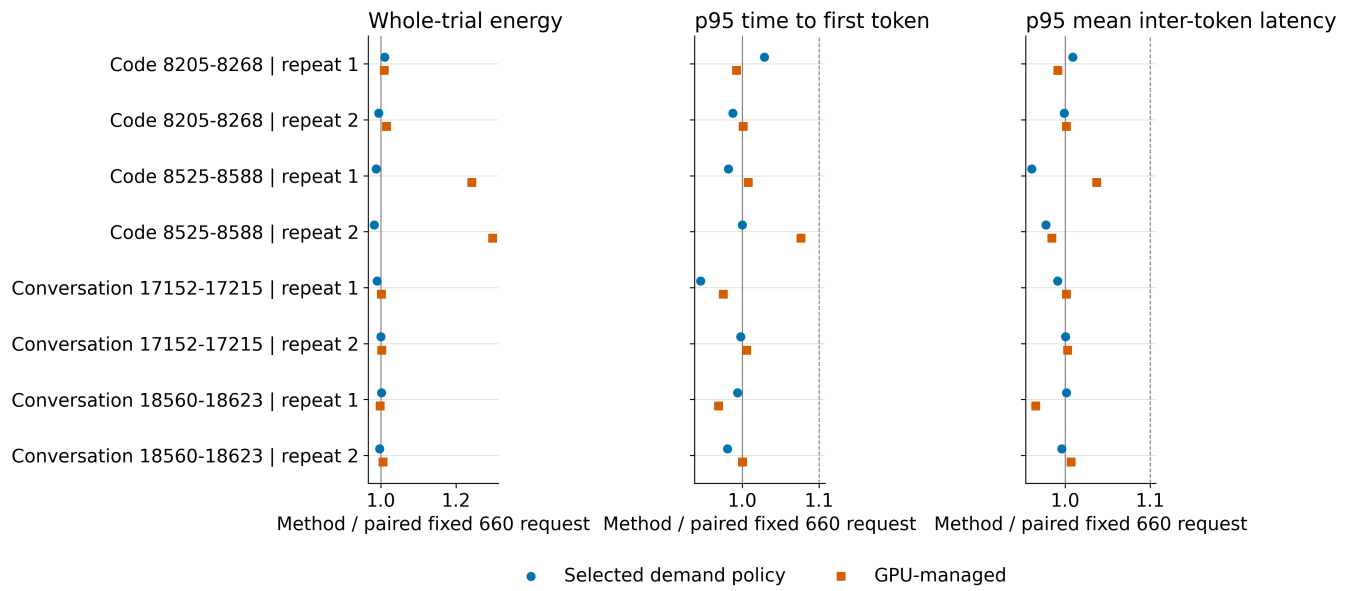
The adaptive heuristic requests 660 MHz when requests await their first token or more than three requests stream output. After a two-second hold under lower demand, it can request 210 MHz. It uses only observations available at the decision time, without future arrivals or exact unfinished output lengths. Existing governor and MPC implementations remain separate; this experiment does not fill their calibration requirements. Both tested development heuristics missed their latency criteria, so the selected demand policy entered held-out evaluation with its exploratory status frozen.

Every held-out adaptive comparison stayed within the allowed 10% latency margin relative to its fixed trial. The largest p95 first-token and mean inter-token latency ratios were 1.029 and 1.009. However, all three methods missed the absolute first-token target on the shorter code block in both repetitions. A favorable paired ratio does not repair an absolute-target failure or retroactively qualify a development policy. These results establish a completed descriptive comparison, not a latency-qualified adaptive-savings claim.

The long code block exposes where the baseline difference arose. Its arrival schedule spans 791.50 seconds and contains a 652.27-second gap between arrivals. Both repetitions preserve the entire gap and final drainage. GPU-managed graphics returned to a high clock while no client request was outstanding. Across this block's two repetitions, GPU-managed energy was 27.04% higher than fixed 660 MHz; adaptive energy was only 1.57% lower than fixed. The shorter code block slightly reversed the adaptive result, using 0.16% more energy than fixed. The aggregate consequently describes this particular workload mix.

Temperature admission bounded each start without matching temperatures or establishing equilibrium. Trials used different allocations and a changed method order between repetitions; the report includes those covariates. A development policy that never switched was slower than an older fixed trial on another allocation, illustrating why such differences cannot be assigned to switching alone. Four blocks with two repeats do not support a population confidence interval, and individual requests are not independent experimental repeats.

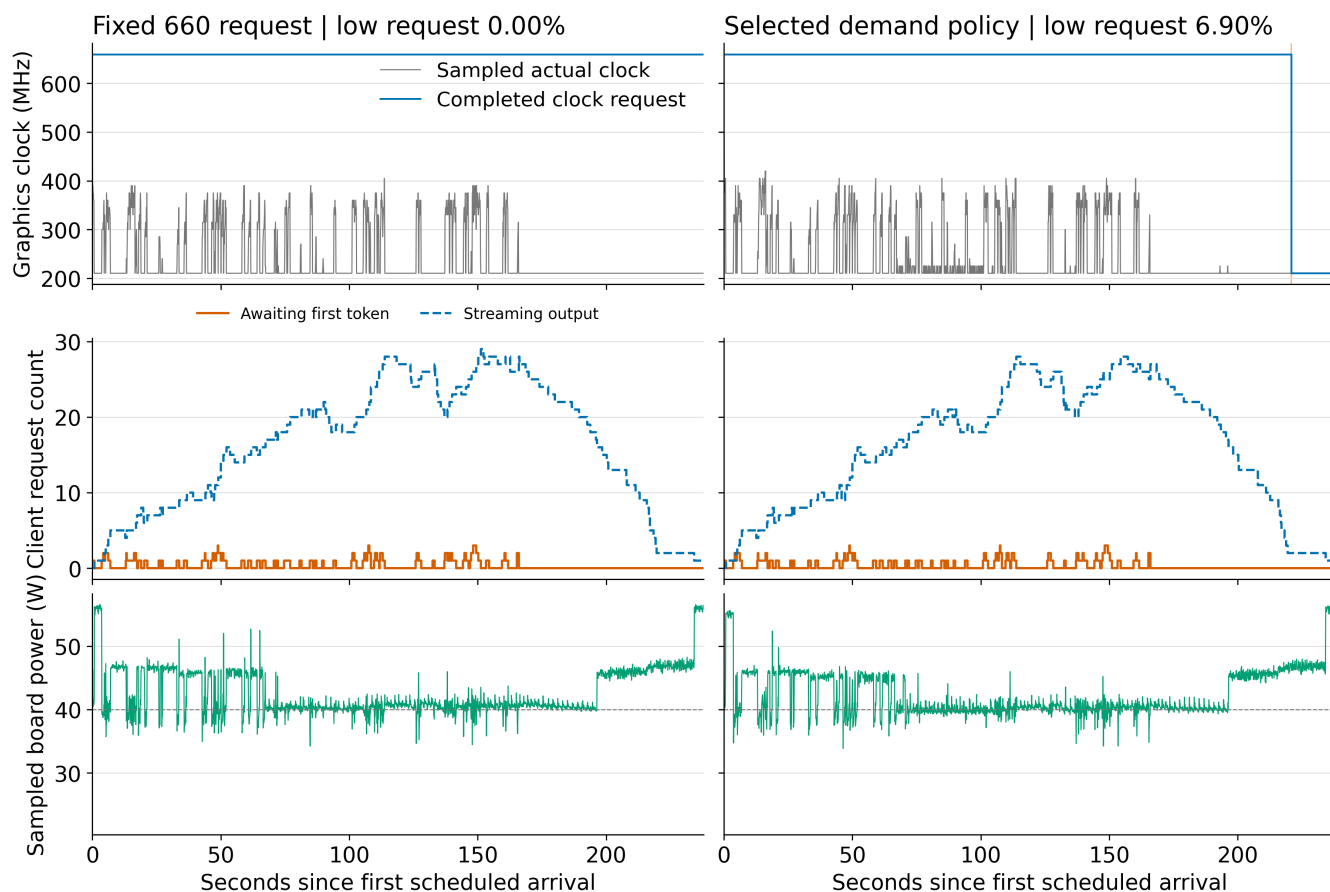
Energy includes all reported GPU power through final completion, including in-trial clock commands. It excludes preparation, cooldown and host energy. Some sensor readings exceeded the configured power limit, including sustained singleton-generation readings near 55-56 W; the cause remains unresolved without a physical meter. Values are preserved without clipping or rescaling. For this model, hardware regime and replay mix, selecting a moderate fixed clock delivered the substantial measured benefit; the tested adaptive heuristic added a small, workload-dependent increment. These observations do not rule out gains from other policies or hardware.



Eight groups are four unique blocks with two repetitions. Each point divides an independently valid method trial by its fixed trial in the same acquisition. Solid lines mark ratio 1; dashed latency lines mark the 1.1 relative limit. Rings flag a latency failure. Blank annotations retain invalid outcomes. Absolute targets and exploratory selection are separate. No population confidence interval is implied.



Code 8205-8268 | repeat 1. The first complete triplet in frozen order is selected without effect-size ranking. Requested-clock steps begin at recorded command completion; narrow shaded intervals mark command execution. No-switch traces are retained. Actual clocks and power are samples, not proof of continuous frequency or exact GPU phases. The dashed 40 W line is the configured limit; sensor readings may exceed it.



Conversation 17152-17215 | repeat 1. The first complete triplet in frozen order is selected without effect-size ranking. Requested-clock steps begin at recorded command completion; narrow shaded intervals mark command execution. No-switch traces are retained. Actual clocks and power are samples, not proof of continuous frequency or exact GPU phases. The dashed 40 W line is the configured limit; sensor readings may exceed it.

Energy and latency by source and block

Energy saved is 1 - summed method joules / summed paired fixed joules, restricted to the reported valid groups. Negative values mean more energy. Four blocks and two repetitions are descriptive experimental units.

Scope	Group	Method	Valid groups	Energy saved	Max TTFT ratio	Max ITL ratio
all	all	Selected demand policy	8/8	0.77%	1.029	1.009
source	code	Selected demand policy	4/4	1.17%	1.029	1.009
source	conversation	Selected demand policy	4/4	0.29%	0.998	1.002
block	code 8205-8268/rate_2x	Selected demand policy	2/2	-0.16%	1.029	1.009
block	code 8525-8588/rate_2x	Selected demand policy	2/2	1.57%	1.000	0.978
block	conversation 17152-17215/rate_4x	Selected demand policy	2/2	0.55%	0.998	1.001
block	conversation 18560-18623/rate_4x	Selected demand policy	2/2	0.09%	0.994	1.002
all	all	GPU-managed	8/8	-11.50%	1.077	1.037
source	code	GPU-managed	4/4	-21.00%	1.077	1.037
source	conversation	GPU-managed	4/4	-0.18%	1.006	1.007
block	code 8205-8268/rate_2x	GPU-managed	2/2	-1.21%	1.001	1.002
block	code 8525-8588/rate_2x	GPU-managed	2/2	-27.04%	1.077	1.037
block	conversation 17152-17215/rate_4x	GPU-managed	2/2	-0.21%	1.006	1.003
block	conversation 18560-18623/rate_4x	GPU-managed	2/2	-0.15%	1.000	1.007

Absolute source-relative targets

Targets were frozen from development measurements. A relative paired improvement can still fail an absolute target. The adaptive candidate was not latency-qualified in development.

Group	Method	Energy (J)	TTFT (s)	Target (s)	TTFT	Mean ITL (s)	Target (s)	ITL
Code 8205-8268 repeat 1	Fixed 660 request	6172.6	31.182	29.183	fail	1.538	1.739	pass
Code 8205-8268 repeat 1	Selected demand policy	6234.1	32.082	29.183	fail	1.551	1.739	pass
Code 8205-8268 repeat 1	GPU-managed	6230.5	30.946	29.183	fail	1.525	1.739	pass
Code 8525-8588 repeat 1	Fixed 660 request	20581.2	10.583	29.183	pass	1.249	1.739	pass
Code 8525-8588 repeat 1	Selected demand policy	20307.3	10.389	29.183	pass	1.200	1.739	pass
Code 8525-8588 repeat 1	GPU-managed	25574.6	10.666	29.183	pass	1.296	1.739	pass
Conversation 17152-17215 repeat 1	Fixed 660 request	10150.2	2.044	3.694	pass	0.249	0.383	pass
Conversation 17152-17215 repeat 1	Selected demand policy	10045.1	1.933	3.694	pass	0.247	0.383	pass
Conversation 17152-17215 repeat 1	GPU-managed	10168.7	1.993	3.694	pass	0.250	0.383	pass
Conversation 18560-18623 repeat 1	Fixed 660 request	13125.7	2.832	3.694	pass	0.207	0.383	pass
Conversation 18560-18623 repeat 1	Selected demand policy	13149.8	2.814	3.694	pass	0.207	0.383	pass
Conversation 18560-18623 repeat 1	GPU-managed	13097.5	2.744	3.694	pass	0.200	0.383	pass
Conversation 18560-18623 repeat 2	Fixed 660 request	12555.4	2.435	3.694	pass	0.174	0.383	pass
Conversation 18560-18623 repeat 2	Selected demand policy	12507.4	2.388	3.694	pass	0.173	0.383	pass
Conversation 18560-18623 repeat 2	GPU-managed	12623.2	2.436	3.694	pass	0.175	0.383	pass
Conversation 17152-17215 repeat 2	Fixed 660 request	9684.3	1.739	3.694	pass	0.229	0.383	pass
Conversation 17152-17215 repeat 2	Selected demand policy	9680.4	1.735	3.694	pass	0.229	0.383	pass
Conversation 17152-17215 repeat 2	GPU-managed	9706.7	1.749	3.694	pass	0.230	0.383	pass
Code 8525-8588 repeat 2	Fixed 660 request	20978.2	10.758	29.183	pass	1.316	1.739	pass
Code 8525-8588 repeat 2	Selected demand policy	20597.9	10.757	29.183	pass	1.287	1.739	pass
Code 8525-8588 repeat 2	GPU-managed	27223.2	11.582	29.183	pass	1.296	1.739	pass
Code 8205-8268 repeat 2	Fixed 660 request	6515.2	35.617	29.183	fail	1.648	1.739	pass
Code 8205-8268 repeat 2	Selected demand policy	6474.5	35.176	29.183	fail	1.647	1.739	pass
Code 8205-8268 repeat 2	GPU-managed	6610.6	35.662	29.183	fail	1.651	1.739	pass

Client activity and long replay exposure

Client-active means a dispatched request has not completed. Its complement is not an exact GPU-idle classification. Completion changes can change partition duration; these are descriptive energy partitions, not counterfactual phase savings.

Group	Method	Arrival span (s)	Active (s)	Active (J)	No outstanding (s)	No outstanding (J)
Code 8205-8268 repeat 1	Fixed 660 request	110.0	143.9	5935.1	10.9	237.5
Code 8205-8268 repeat 1	Selected demand policy	110.0	145.5	6005.5	10.5	228.6
Code 8205-8268 repeat 1	GPU-managed	110.0	143.5	5918.5	10.7	312.0
Code 8525-8588 repeat 1	Fixed 660 request	791.5	145.1	6475.9	673.2	14105.2
Code 8525-8588 repeat 1	Selected demand policy	791.5	145.0	6471.5	673.1	13835.7
Code 8525-8588 repeat 1	GPU-managed	791.5	145.7	6564.4	672.8	19010.2
Conversation 17152-17215 repeat 1	Fixed 660 request	165.1	237.9	10150.2	0.0	0.0
Conversation 17152-17215 repeat 1	Selected demand policy	165.1	237.2	10045.1	0.0	0.0
Conversation 17152-17215 repeat 1	GPU-managed	165.1	237.9	10168.7	0.0	0.0
Conversation 18560-18623 repeat 1	Fixed 660 request	219.7	291.2	13125.7	0.0	0.0
Conversation 18560-18623 repeat 1	Selected demand policy	219.7	291.4	13149.8	0.0	0.0
Conversation 18560-18623 repeat 1	GPU-managed	219.7	290.5	13097.5	0.0	0.0
Conversation 18560-18623 repeat 2	Fixed 660 request	219.7	284.5	12555.4	0.0	0.0
Conversation 18560-18623 repeat 2	Selected demand policy	219.7	284.4	12507.4	0.0	0.0
Conversation 18560-18623 repeat 2	GPU-managed	219.7	284.9	12623.2	0.0	0.0
Conversation 17152-17215 repeat 2	Fixed 660 request	165.1	231.7	9684.3	0.0	0.0
Conversation 17152-17215 repeat 2	Selected demand policy	165.1	231.8	9680.4	0.0	0.0
Conversation 17152-17215 repeat 2	GPU-managed	165.1	232.0	9706.7	0.0	0.0
Code 8525-8588 repeat 2	Fixed 660 request	791.5	146.7	6609.0	672.0	14369.3
Code 8525-8588 repeat 2	Selected demand policy	791.5	146.6	6556.9	672.2	14041.0
Code 8525-8588 repeat 2	GPU-managed	791.5	148.6	6671.3	671.9	20551.9
Code 8205-8268 repeat 2	Fixed 660 request	110.0	152.3	6295.6	9.6	219.6
Code 8205-8268 repeat 2	Selected demand policy	110.0	151.6	6254.6	9.8	219.8
Code 8205-8268 repeat 2	GPU-managed	110.0	152.6	6312.5	9.6	298.1

Direct adaptive / GPU-managed comparison

Secondary observed comparison. It does not replace the frozen fixed-relative selection or latency criteria.

Group	Energy ratio	TTFT ratio	Mean ITL ratio
Code 8205-8268 repeat 1	1.001	1.037	1.017
Code 8525-8588 repeat 1	0.794	0.974	0.926
Conversation 17152-17215 repeat 1	0.988	0.969	0.990
Conversation 18560-18623 repeat 1	1.004	1.026	1.037
Conversation 18560-18623 repeat 2	0.991	0.980	0.989
Conversation 17152-17215 repeat 2	0.997	0.992	0.998
Code 8525-8588 repeat 2	0.757	0.929	0.993
Code 8205-8268 repeat 2	0.979	0.986	0.997

Temperature and execution-order sensitivity

Descriptive subsets from recorded admission temperatures and order. They are not randomized temperature interventions or independent estimates.

Method	Subset	Groups	Energy saved
Selected demand policy	cooler or equal start	5	0.87%
Selected demand policy	method after fixed	4	0.60%
Selected demand policy	method before fixed	4	0.94%
Selected demand policy	warmer start	3	0.59%
GPU-managed	cooler or equal start	4	-12.00%
GPU-managed	method after fixed	4	-12.48%
GPU-managed	method before fixed	4	-10.50%
GPU-managed	warmer start	4	-10.93%

Methods and scope

The experiment contains four unique 64-request source blocks and two repetitions, giving eight groups with fixed, selected adaptive, and GPU-managed trials. Code uses `rate_2x` timing; conversation uses the revised `rate_4x` timing. The fixed request is the common development-selected 660 MHz setting. Failed and unstarted outcomes remain in `outcomes.csv`; absent comparisons are not zero savings.

Whole-trial energy integrates the selected NVIDIA board-power field through final request completion. The configuration records 40 W and 6,251 MHz memory. Sensor power above the configured limit remains an unresolved measurement finding, not a physical-meter conclusion. Client awaiting/streaming categories are not exact GPU prefill/decode phases.

Azure contributes lengths and arrival times, not original text. Prompts use synthetic repeated token IDs; sampling uses temperature zero and enforced output lengths, with prefix caching disabled. This does not establish semantic production-workload behavior or exact computational reproducibility.

The admission ceiling remains 62 C over ten seconds with at least 9.5 seconds of coverage. The later 900-second allowance changes waiting time only; older 180-second failures remain recorded. The separate 73 C development gate and 77/79 C watchdogs are not admission thresholds or evidence of equilibrium.

The selected causal heuristic is exploratory because it failed development latency qualification. Governor/MPC calibration is not established by these trials. The results do not rule out other phase-aware policies, voltage effects, or different hardware. Differences across allocations cannot be attributed to switching alone; paired held-out comparisons are the relevant observed contrasts.

Trace attribution

The code and conversation trace metadata are adapted from the [Azure LLM Inference Dataset 2023](#), released under [CC BY 4.0](#). Credit: Pratyush Patel, Esha Choukse, Chaojie Zhang, Aashaka Shah, Inigo Goiri, Saeed Maleki, and Ricardo Bianchini, [Splitwise: Efficient generative LLM inference using phase splitting](#), ISCA 2024. We selected 64-request blocks and rescaled arrival timing. We replayed the recorded input/output token lengths using synthetic repeated-token-ID prompts; the traces do not contain the original prompt text.

Audit SHA-256: f4db4c8ba8a4cc96e83d3f9a2a2d38c5934035799ecdd1d24aa49d50cf1d61a2